Vamsi Krishna Sai Kavety

HIL/SIL TEST ENGINEER | EMBEDDED SOFTWARE INTEGRATION

Skills

Test & Validation:
HIL rigs (Vector VT System, RT Rack), Fault Insertion Unit (FIU), CANoe/CANalyzer, LabVIEW TestStand

Embedded Programming:
C/C++, Python, Shell

Automotive Protocols:
UART, CAN, RS-232, RS-485, UDS

AUTOSAR & Config Tools:
DaVinci, EB Tresos, ARXML files

CI/CD & Test Automation:
Jenkins, Git/Gerrit/GitLab, Jira, Continental ATP/TEMPPO

DevOps & Cloud:
Docker, Kubernetes, Terraform, Ansible, AWS, Azure

Debuggers:
Lauterbach/Trace32, JTAG, PE micro MULTILINK, ICAS1 Debugger

Configuration Management:
PTC Integrity, IBM DOORS, MKS

Standards & Process:
ASPICE (HWE.2, HWE.3, SWE.5), ISTQB Foundation, Agile/Scrum

Cross Compilers/IDE:
ARM GCC, GHS, S32 Design Studio, Eclipse, Visual Studio

Contact

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Personal Information

- **Nationality:** German
- **Education:** Master of Science - Automotive Software Engineering — TU Chemnitz, Germany, 2018; PG Program - Data Science & AI (ongoing) — ExcelR, India, 2024-Present; B.Tech - Electrical & Electronics Engineering — SVEC, JNTU, India, 2008
- **Languages:** English - Fluent, German - B1 completed, B2 in progress, Telugu - Mother tongue, Hindi
- **Driving License:** Class B/BE

About Me

- 11+ years across embedded hardware/software testing, HIL verification, and automotive module integration for VW, Daimler, and Fiat/Chrysler OEM programs.
- HIL test engineer: maintained and upgraded Vector VT System/RT Rack rigs, executed Fault Insertion Unit (FIU) testing, and automated regression suites with LabVIEW TestStand and Python.
- Automotive software integrator: Gateway Control Unit (ICAS1/ID.3) and Telematics/Body Control module integration using AUTOSAR (DaVinci, EB Tresos) with Jira/Jenkins CI/CD.
- ASPICE-aligned: implemented HWE.2, HWE.3, SWE.5 process phases; hands-on with Continental ATP/TEMPPO test management framework.
- Growing DevOps/AI skill set: Docker, Kubernetes, Terraform, Ansible, AWS/Azure CI/CD, currently pursuing a PG program in Data Science & AI.

Core Competencies

- HIL/SIL Test Engineering: Vector VT System, RT Rack, CANoe/CANalyzer, LabVIEW TestStand, Fault Insertion Unit (FIU), root cause analysis.
- Automotive Software Integration: Gateway/Telematics/Body/Powertrain ECUs, AUTOSAR (DaVinci, EB Tresos, ARXML), software flashing & diagnostics (ODIS).
- CI/CD & Test Automation: Jenkins, Git/Gerrit/GitLab, Jira, Python-based regression and fault-injection scripting, Continental ATP/TEMPPO.
- Debugging & Diagnostics: Lauterbach/Trace32, JTAG, PE Micro Multilink debuggers, CAN/UART/RS-232/RS-485 protocol analysis.
- DevOps & Cloud (growing): Docker, Kubernetes, Terraform, Ansible, AWS, Azure.
- Standards & Process: ASPICE (HWE.2, HWE.3, SWE.5), ISTQB Foundation, Agile/Scrum, IBM DOORS, PTC Integrity/MKS configuration management.

Certifications

- ISTQB Certified Tester - Foundation Level (CTFL)

Current Role Focus

- HIL/SIL Test Engineer - Automotive ECU & Gateway Platforms
- Software Integration Engineer - AUTOSAR/Gateway/Telematics Systems
- DevOps-Aligned Test Automation Engineer - CI/CD & Cloud Tooling

Work Experience

Software Engineer/Test Engineer - HIL (VW)

Manu Dextra / Vitesco Technologies - Regensburg

Jul 2021 - Mar 2024

- Maintained and upgraded HIL test rigs (Vector VT System, RT Rack) across multiple ECU programs, minimizing test bench downtime through advanced hardware/software troubleshooting.
- Designed and executed fault injection tests using the Fault Insertion Unit (FIU) to simulate electrical failures and verify ECU diagnostic and fault-handling behavior.
- Automated regression and validation suites using LabVIEW TestStand and Python scripts for signal simulation and fault injection.
- Executed and validated software downloads (SWDL) on EMS ECUs, coordinating with development and test teams to ensure test environment readiness.
- Managed hardware inventory, calibration schedules, and system documentation; contributed to continuous improvement of HIL methodologies and tooling.

Software Engineer - Vehicle Testing

Manu Dextra / Continental Automotive - Regensburg

Aug 2019 - Jun 2021

- Conducted manual and automated in-vehicle/lab testing on the VW ID.3 platform to validate Gateway Control Unit (ICAS1) requirements for every sprint release.
- Owned software module integration within an Agile team (Jira) for the VW Gateway Control Unit, coordinating with cross-site development teams.
- Established and managed Jenkins-based CI/CD integration testing setups, improving release traceability and regression coverage.
- Performed software flashing and diagnostics on hardware using ICAS1 debuggers, VW ODIS tools, and custom Python scripts.

Software Engineer/Test Engineer - ECU Verification

IPN Brainpower / Continental Engineering Services - Nurnberg

Feb 2018 - Apr 2019

- Executed automated and manual HIL test procedures covering nominal, boundary, and complex powertrain control scenarios.
- Designed fault injection tests (FIU) to simulate electrical failures and verify diagnostic/fault-handling behavior of ECUs.
- Performed data analysis and root cause analysis (RCA), reporting defects via Jira for software/hardware anomalies.

Master Thesis Candidate

Conti Temic Microelectronics GmbH (Body & Security) - Markdorf

Jul 2017 - Dec 2017

- Researched Over-the-Air (OTA) software/firmware update delivery mechanisms for automotive service center systems.

Student Intern - Testing

ADC Automotive Distance Control Systems GmbH - Lindau/Ulm

Jan 2017 - Jun 2017

- Developed Python scripts to automate repetitive vehicle component testing and validation tasks in line with automotive standards.

Senior Hardware Engineer / Senior HIL Test Engineer - Power Train

Robert Bosch Engineering - Coimbatore, India

May 2011 - Mar 2014

- Led integration and validation of ECUs for diesel/gasoline powertrains, coordinating with Bosch GmbH and OEMs (Fiat/Chrysler) on hardware specifications.
- Conducted functional, EMC, and environmental testing per automotive quality standards; performed root cause analysis for system integration issues.

Engineer R&D

McML Protection Technologies - Hyderabad, India

Jul 2009 - Apr 2011

- Designed proof-of-concept hardware prototypes and developed/validated embedded C/C++ firmware and drivers.
- Set up test benches and performed unit/integration/system-level testing and debugging of embedded hardware/software.